B1 Flows, Fluctuations and Complexity — Model Solutions, 2018

Question 1 — Bernoulli and wind over ripples

(a) Bernoulli's theorem

Problem. Derive the conserved quantity along streamlines of a steady, inviscid, incompressible, irrotational flow under gravity from Navier–Stokes; identify each term; comment on the compressible case.

Drop $\nu\nabla^2\mathbf{u}$ (inviscid) and $\partial_t\mathbf{u}$ (steady) from N–S:

$$(\mathbf{u}\cdot\nabla)\mathbf{u} + \tfrac{1}{\rho}\nabla p + g\hat{\mathbf{k}} = 0.$$

Apply identity $(\mathbf{u}\cdot\nabla)\mathbf{u} = \nabla(|\mathbf{u}|^2/2) - \mathbf{u}\times(\nabla\times\mathbf{u})$:

$$\nabla\left(\tfrac{|\mathbf{u}|^2}{2}\right) - \mathbf{u}\times\boldsymbol{\omega} + \tfrac{1}{\rho}\nabla p + g\hat{\mathbf{k}} = 0, \qquad \boldsymbol{\omega} = \nabla\times\mathbf{u}.$$

Irrotational flow, $\boldsymbol{\omega} = 0$:

$$\nabla\left(\tfrac{|\mathbf{u}|^2}{2}\right) + \tfrac{1}{\rho}\nabla p + g\hat{\mathbf{k}} = 0.$$

Write $g\hat{\mathbf{k}} = \nabla(gz)$ and incompressibility $\rho = \text{const}$ allows $\nabla p/\rho = \nabla(p/\rho)$:

$$\nabla\left(\tfrac{|\mathbf{u}|^2}{2} + \tfrac{p}{\rho} + gz\right) = 0.$$

Integrate (since the gradient of a scalar is zero implies the scalar is constant):

$$\boxed{\tfrac{1}{2}|\mathbf{u}|^2 + \tfrac{p}{\rho} + gz = \text{const everywhere}}$$

Origin of terms: $\frac{1}{2}|\mathbf{u}|^2$ is kinetic energy per unit mass (advective term); p/ρ is the work done by pressure forces per unit mass (pressure term); gz is gravitational potential energy per unit mass.

Compressible interpretation: with $\rho = \rho(p)$, $\nabla p/\rho$ is no longer $\nabla(p/\rho)$. The pressure term must be replaced by the enthalpy $\int \mathrm{d}p/\rho(p)$, which becomes c_pT for an ideal gas. Bernoulli then expresses conservation of total enthalpy along a streamline.

(b) Air flow over rippled water

Problem. Justify irrotationality; show $u = U_0[1 + a\sin(2\pi x/\lambda)e^{-bz}]$ is consistent with incompressibility and the BCs above $h = \eta\sin(2\pi x/\lambda)$; find a, b; sketch streamlines.

Irrotationality: at $x < 0$ the flow is uniform, hence $\boldsymbol{\omega} = 0$. With $\nu_{\text{air}} \to 0$, the vorticity equation $\mathrm{D}\boldsymbol{\omega}/\mathrm{D}t = (\boldsymbol{\omega}\cdot\nabla)\mathbf{u}$ has no source, so $\boldsymbol{\omega} = 0$ is preserved along streamlines downstream of $x = 0$.

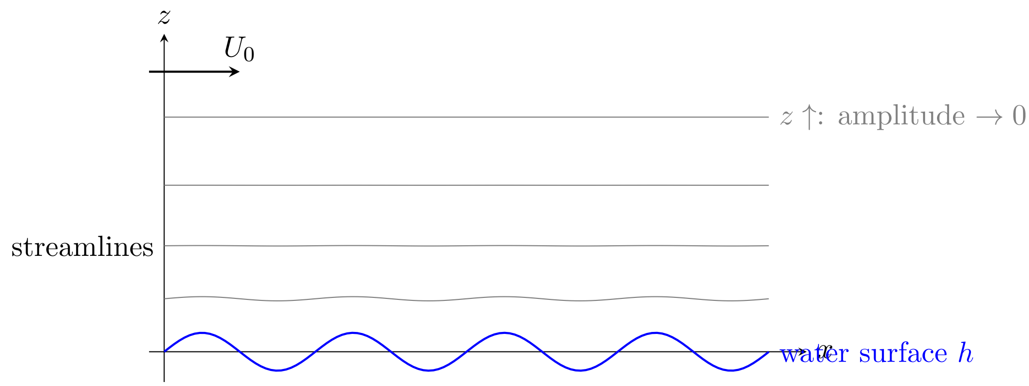
Let $k = 2\pi/\lambda$. Incompressibility in 2D:

$$\partial_x u + \partial_z w = 0.$$

Compute $\partial_x u$:
$$\partial_x u = U_0 ak\cos(kx)\, e^{-bz}.$$
Rearrange for $\partial_z w$:
$$\partial_z w = -U_0 ak\cos(kx)\, e^{-bz}.$$
Integrate in z:
$$w = \tfrac{U_0 ak}{b}\cos(kx)\, e^{-bz} + C(x).$$
Far-field BC $w \to 0$ as $z \to \infty$ forces $C(x) = 0$:
$$w(x,z) = \tfrac{U_0 ak}{b}\cos(kx)\, e^{-bz}.$$
Kinematic BC at $z = 0$ (linearised, since $\eta \ll \lambda$ and phase speed $\ll U_0$, hence quasi-steady):
$$w(x,0) = U_0\, \partial_x h.$$
Compute $\partial_x h$:
$$\partial_x h = \eta k\, \cos(kx).$$
Substitute:
$$w(x,0) = U_0\, \eta k\, \cos(kx).$$
Equate to expression above evaluated at $z = 0$:
$$\tfrac{U_0 ak}{b}\cos(kx) = U_0\, \eta k\, \cos(kx).$$
Cancel $U_0 k\cos(kx)$:
$$\tfrac{a}{b} = \eta.$$
Solve for a:
$$a = \eta b. \tag{i}$$
Compute $\partial_z u$:
$$\partial_z u = -U_0 ab\sin(kx)\, e^{-bz}.$$
Compute $\partial_x w$:
$$\partial_x w = -\tfrac{U_0 ak^2}{b}\sin(kx)\, e^{-bz}.$$
Impose irrotationality $\partial_z u - \partial_x w = 0$:
$$-U_0 ab\sin(kx)e^{-bz} + \tfrac{U_0 ak^2}{b}\sin(kx)e^{-bz} = 0.$$
Cancel common factor $U_0 a\sin(kx)e^{-bz}$:
$$-b + \tfrac{k^2}{b} = 0.$$
Multiply through by b:
$$b^2 = k^2.$$
Take positive root (decay as $z \to \infty$):
$$b = k = \tfrac{2\pi}{\lambda}.$$
Substitute into (i):
$$a = \eta k.$$
Boxed:
$$\boxed{a = \eta k = \tfrac{2\pi\eta}{\lambda}, \qquad b = k = \tfrac{2\pi}{\lambda}}$$

Boundary conditions used: (1) $w \to 0$ as $z \to \infty$ (undisturbed flow far above); (2) $w(x,0) = U_0\, \partial_x h$ (kinematic, linearised); (3) $u \to U_0$ as $z \to \infty$.

Streamline sketch:



Streamlines mirror the surface near $z = 0$ and flatten exponentially with decay length $1/k = \lambda/(2\pi)$.

(c) Pressure at surface; ripple growth

Problem. Find p at $z = 0$; explain how a steady wind grows ripples even with inviscid air; does the mechanism depend on direction of propagation; two amplitude-limiting processes.

Bernoulli on surface streamline ($z \approx 0$, drop gz since gravitational term is constant along the surface to linear order):

$$p + \tfrac{1}{2}\rho_a|\mathbf{u}|^2 = p_\infty + \tfrac{1}{2}\rho_a U_0^2.$$

Evaluate u at $z = 0$ (with $b = k$, $a = \eta k$):

$$u(x, 0) = U_0[1 + a\sin(kx)].$$

Evaluate w at $z = 0$ (with $b = k$):

$$w(x, 0) = \tfrac{U_0 ak}{k}\cos(kx) = U_0 a\cos(kx).$$

Form $|\mathbf{u}|^2 = u^2 + w^2$ on the surface:

$$|\mathbf{u}|^2 = U_0^2\big[(1 + a\sin kx)^2 + a^2\cos^2 kx\big].$$

Expand the square:

$$|\mathbf{u}|^2 = U_0^2\big[1 + 2a\sin(kx) + a^2\sin^2(kx) + a^2\cos^2(kx)\big].$$

Apply $\sin^2 + \cos^2 = 1$:

$$|\mathbf{u}|^2 = U_0^2\big[1 + 2a\sin(kx) + a^2\big].$$

Linearise in $a = \eta k \ll 1$, drop $O(a^2)$:

$$|\mathbf{u}|^2 \approx U_0^2\big[1 + 2a\sin(kx)\big].$$

Substitute into Bernoulli:

$$p(x, 0) = p_\infty + \tfrac{1}{2}\rho_a U_0^2 - \tfrac{1}{2}\rho_a U_0^2[1 + 2a\sin(kx)].$$

Cancel the $\frac{1}{2}\rho_a U_0^2$ terms:

$$p(x, 0) = p_\infty - \rho_a U_0^2\, a\sin(kx).$$

Insert $a = \eta k$:

$$\boxed{p(x,0) = p_\infty - \rho_a U_0^2\, \eta k\, \sin(kx)}$$

Phase relation: surface elevation is $h = \eta \sin(kx)$; pressure perturbation is $\delta p = -\rho_a U_0^2 \eta k \sin(kx)$. Pressure is *lowest at the crests* ($\sin kx = +1$) and *highest at the troughs* ($\sin kx = -1$). This pressure pattern is in antiphase with the displacement, which alone would not transfer energy (energy transfer requires a pressure component in phase with the surface velocity).

For energy transfer, allow ripples to propagate with phase speed $c \ll U_0$: $h = \eta \sin(k(x - ct))$. The kinematic BC then becomes $w(x,0) = (U_0 - c)\, \partial_x h$, so the pressure perturbation acquires a small cos-component proportional to c/U_0 that is in phase with $\partial_t h$. The wind does positive work on the surface, transferring energy from the air to the water and amplifying the ripple. This is the inviscid Miles–type instability mechanism.

Direction dependence: yes. Reversing the propagation direction ($c \to -c$) flips the sign of the work integral $\oint p\, \partial_t h\, \mathrm{d}x$, so wind amplifies only ripples propagating in the same sense as U_0; counter-propagating ripples are damped.

Amplitude-limiting processes (any two): (1) viscous dissipation in the water (and the thin air boundary layer); (2) nonlinear wave breaking / whitecapping once $\eta k \sim 1$; (3) energy transfer by nonlinear wave–wave interactions to other wavenumbers; (4) surface tension increasing the restoring force as k grows.

Question 2 — Pipe flow

(a) Definitions

Problem. Define steady, axisymmetric, fully developed flow.

Steady: $\partial_t \mathbf{u} = 0$. Axisymmetric: $\partial_\Theta \mathbf{u} = 0$, with $u_\Theta = 0$ (no swirl). Fully developed: $\partial_z \mathbf{u} = 0$, so the velocity profile is independent of axial position; mass conservation then forces $u_r = 0$ and the only non-zero component is $u_z(r)$.

(b) Hagen–Poiseuille profile

Problem. State the conditions for the given axial momentum equation; integrate; apply BCs to find A, B, C; obtain flow rate vs ΔP.

Conditions: incompressible, Newtonian, steady, axisymmetric, fully-developed flow with no swirl and no body force in z. Then the z component of N–S reduces to the cited equation, with $\partial_z p$ a constant (since LHS depends only on r).

Multiply both sides by r/ν:

$$\tfrac{r}{\rho\nu}\, \partial_z p = \frac{\partial}{\partial r}(r\, \partial_r u_z)\,.$$

Integrate once in r (treating $\partial_z p$ as constant):

$$r\, \partial_r u_z = \tfrac{r^2}{2\rho\nu}\, \partial_z p + B.$$

Divide by r:

$$\partial_r u_z = \tfrac{r}{2\rho\nu}\, \partial_z p + \tfrac{B}{r}.$$

Integrate again in r:

$$u_z = \tfrac{r^2}{4\rho\nu}\, \partial_z p + B \ln r + C.$$

Read off the prefactor:

$$A = \tfrac{1}{4\rho\nu} = \tfrac{1}{4\mu}.$$

BC1 (regularity at $r=0$): u_z finite forces $B=0$.

BC2 (no-slip at $r=R$): $u_z(R)=0$:

$$0 = A\,\partial_z p\,R^2 + C.$$

Solve for C:

$$C = -A\,\partial_z p\,R^2.$$

Substitute A, B, C back:

$$\boxed{u_z(r) = \tfrac{1}{4\mu}\,\partial_z p\,(r^2 - R^2)}$$

Use $\partial_z p = -\Delta P/L$ (pressure drops along $+z$):

$$u_z(r) = -\tfrac{\Delta P}{4\mu L}(r^2 - R^2).$$

Flip sign:

$$u_z(r) = \tfrac{\Delta P}{4\mu L}(R^2 - r^2).$$

Volumetric flow rate through annulus $2\pi r\,\mathrm{d}r$:

$$Q = \int_0^R u_z(r)\,2\pi r\,\mathrm{d}r.$$

Substitute u_z:

$$Q = \tfrac{2\pi\Delta P}{4\mu L}\int_0^R (R^2 r - r^3)\,\mathrm{d}r.$$

Evaluate the integral:

$$\int_0^R (R^2 r - r^3)\,\mathrm{d}r = \left[\tfrac{R^2 r^2}{2} - \tfrac{r^4}{4}\right]_0^R = \tfrac{R^4}{2} - \tfrac{R^4}{4} = \tfrac{R^4}{4}.$$

Substitute back:

$$Q = \tfrac{\pi\Delta P}{2\mu L}\cdot\tfrac{R^4}{4}.$$

Simplify:

$$\boxed{Q = \tfrac{\pi R^4\,\Delta P}{8\mu L}}$$

(c) Log–log Q vs ΔP

Problem. Sketch $\log Q$ vs $\log\Delta P$ for oil through $L = 1\,\mathrm{m}$, $R = 1\,\mathrm{cm}$, $\nu_o = 10^{-5}\,\mathrm{m^2 s^{-1}}$, $\rho_o = 10^3\,\mathrm{kg m^{-3}}$, over $\Delta P \in [0.1\,\mathrm{Pa}, 10^6\,\mathrm{Pa}]$; show $u_z(r)$ shapes for each regime.

Compute dynamic viscosity:

$$\mu = \rho\nu = (10^3)(10^{-5}) = 10^{-2}\ \mathrm{Pa\,s}.$$

Substitute into Hagen–Poiseuille with $R = 10^{-2}$ m, $L = 1$ m:

$$Q = \tfrac{\pi(10^{-2})^4}{8(10^{-2})(1)}\,\Delta P.$$

Simplify the prefactor:

$$Q = \tfrac{\pi}{8}\times 10^{-6}\,\Delta P\ \ [\mathrm{m^3 s^{-1}}].$$

Numerically:

$$Q_{\mathrm{lam}} \approx 3.93\times 10^{-7}\,\Delta P.$$

Define mean speed $\bar{u} = Q/(\pi R^2)$. Reynolds number:

$$Re = \tfrac{2R\bar{u}}{\nu} = \tfrac{2Q}{\pi R\nu}.$$

Set $Re = Re_c \approx 2300$ and solve for critical flow rate:

$$Q_c = \tfrac{\pi R\nu\, Re_c}{2}.$$

Insert numbers:

$$Q_c = \tfrac{\pi(10^{-2})(10^{-5})(2300)}{2}.$$

Evaluate:

$$Q_c \approx 3.6\times10^{-4}\,\mathrm{m^3s^{-1}}.$$

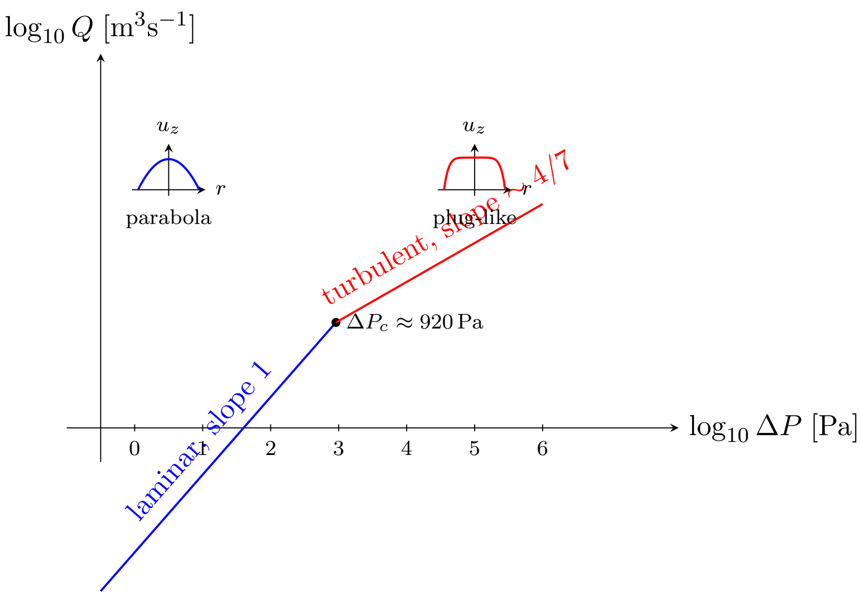
Critical pressure drop from laminar relation:

$$\Delta P_c \approx \tfrac{Q_c}{3.93\times10^{-7}}.$$

Evaluate:

$$\Delta P_c \approx 920\,\mathrm{Pa}.$$

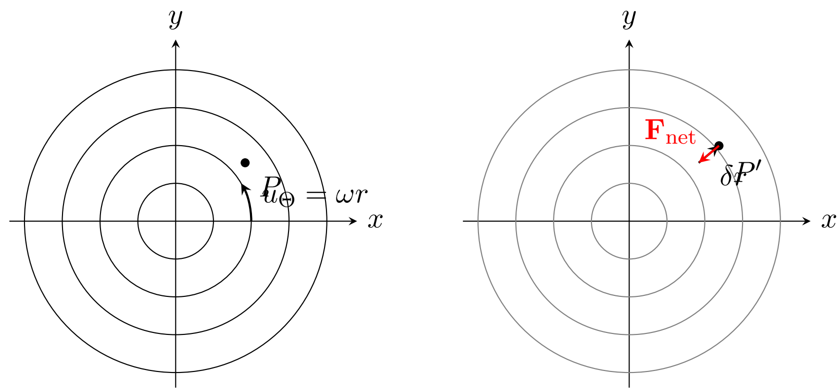
Above ΔP_c flow becomes turbulent; empirically (Blasius friction $f \sim Re^{-1/4}$) $Q \propto \Delta P^{4/7}$, so the log–log slope drops from 1 to $\approx 4/7$.



(d) Swirl and centrifugal restoring force

Problem. Solid-body rotation $u_\Theta = \omega r$. Sketch (r,Θ) streamlines; sketch them after a parcel is displaced by δr; show the parcel feels a body force; deduce why swirl can raise Q.

Streamlines in (r,Θ): concentric circles centred on the axis (same r, all Θ).



Conservation of angular momentum of the parcel during slow radial displacement:

$$L = m\, u_\Theta\, r = \text{const.}$$

Initial value at radius r:

$$L_0 = m\,\omega r \cdot r = m\omega r^2.$$

After displacement $r \to r + \delta r$:

$$u'_\Theta(r+\delta r) = \tfrac{L_0/m}{r+\delta r} = \tfrac{\omega r^2}{r+\delta r}.$$

Linearise using $1/(1+\delta r/r) \approx 1 - \delta r/r$:

$$u'_\Theta \approx \omega r\big(1 - \tfrac{\delta r}{r}\big).$$

Simplify:

$$u'_\Theta \approx \omega r - \omega\,\delta r.$$

Background fluid at $r + \delta r$ moves at:

$$u_\Theta^{\text{bg}} = \omega(r+\delta r) = \omega r + \omega\,\delta r.$$

Required centripetal acceleration to remain on the new streamline is $(u_\Theta^{\text{bg}})^2/(r+\delta r)$; parcel can only supply $(u'_\Theta)^2/(r+\delta r)$. Take the difference:

$$\Delta a_r = \tfrac{(u_\Theta^{\text{bg}})^2 - (u'_\Theta)^2}{r+\delta r}.$$

Substitute and keep $r + \delta r \approx r$ in the denominator:

$$\Delta a_r \approx \tfrac{(\omega r + \omega\delta r)^2 - (\omega r - \omega\delta r)^2}{r}.$$

Apply difference of squares $A^2 - B^2 = (A+B)(A-B)$ with $A = \omega r + \omega\delta r$, $B = \omega r - \omega\delta r$:

$$\Delta a_r \approx \tfrac{(2\omega r)(2\omega\delta r)}{r}.$$

Simplify:

$$\Delta a_r \approx 4\omega^2\,\delta r.$$

Hence the parcel experiences a net *inward* restoring acceleration $4\omega^2\,\delta r$ per unit mass: solid-body rotation is centrifugally stable (Rayleigh criterion: $\mathrm{d}(ru_\Theta)^2/\mathrm{d}r > 0$).

Effect on flow rate: the inward restoring force suppresses radial perturbations and therefore delays the onset of turbulence in the pipe, raising the critical ΔP_c. For values of ΔP just above the no-swirl transition (where the laminar parabola gives a higher Q than turbulent flow at the same ΔP), maintaining laminarity by swirl yields a larger Q.

Question 3 — Rössler attractor

(a) Definitions and dimensionality

Problem. Define fixed point and unstable spiral; explain why sustained finite-amplitude aperiodic flow is impossible in a deterministic 2D dynamical system.

Fixed point $\mathbf{x}^*$: $\mathbf{f}(\mathbf{x}^*) = 0$, so trajectories starting there stay there.

Unstable spiral: a fixed point at which the Jacobian has complex-conjugate eigenvalues $\sigma \pm i\Omega$ with $\sigma > 0$; nearby trajectories spiral outwards with growing radius.

Dimensionality argument: in 2D, deterministic trajectories cannot cross (uniqueness of solutions of ODEs). The Poincaré–Bendixson theorem then forces any bounded trajectory to limit on a fixed point or a closed orbit (limit cycle) — both periodic. Aperiodic motion needs a third dimension so trajectories can go "round the back" without intersecting.

(b) Fixed points and stability

Problem. Find fixed points of Rössler; in the limit $c \gg 1$, $c^2 \gg a^2$, classify the inner fixed point as a function of a.

Fixed point condition: set $\dot{x} = \dot{y} = \dot{z} = 0$:

$$-y - z = 0, \qquad x + ay = 0, \qquad a + z(x - c) = 0.$$

From the first equation:

$$z = -y.$$

From the second equation:

$$x = -ay.$$

Substitute both into the third:

$$a + (-y)(-ay - c) = 0.$$

Expand the product:

$$a + ay^2 + cy = 0.$$

Rewrite as a quadratic in y:

$$ay^2 + cy + a = 0.$$

Apply the quadratic formula:

$$y = \tfrac{-c \pm \sqrt{c^2 - 4a^2}}{2a}.$$

Expand $\sqrt{c^2 - 4a^2}$ for $c^2 \gg a^2$:

$$\sqrt{c^2 - 4a^2} \approx c\big(1 - \tfrac{2a^2}{c^2}\big).$$

Inner root (+ branch, cancellation):

$$y_+ \approx \frac{-c + c(1 - 2a^2/c^2)}{2a} = \frac{-2a^2/c}{2a} = -\tfrac{a}{c}.$$

Outer root ($-$ branch):

$$y_- \approx \tfrac{-c - c}{2a} = -\tfrac{c}{a}.$$

Inner fixed point coordinates:

$$\mathbf{x}_+^* \approx \big(\tfrac{a^2}{c},\, -\tfrac{a}{c},\, \tfrac{a}{c}\big).$$

Outer fixed point:

$$\mathbf{x}_-^* \approx \big(c,\, -\tfrac{c}{a},\, \tfrac{c}{a}\big).$$

Boxed:

$$\boxed{\mathbf{x}^*_\pm = \big(-ay_\pm,\, y_\pm,\, -y_\pm\big), \quad y_\pm = \tfrac{-c\pm\sqrt{c^2-4a^2}}{2a}}$$

Jacobian:

$$\mathbf{J} = \begin{pmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ z & 0 & x-c \end{pmatrix}.$$

At inner fixed point, $z = a/c$, $x - c \approx -c$:

$$\mathbf{J}_+ \approx \begin{pmatrix} 0 & -1 & -1 \\ 1 & a & 0 \\ a/c & 0 & -c \end{pmatrix}.$$

Characteristic polynomial $\det(\mathbf{J}_+ - \lambda\mathbf{I}) = 0$, expanded along the first row:

$$-\lambda\big[(a-\lambda)(-c-\lambda) - 0\big] + 1\big[1\cdot(-c-\lambda) - 0\big] + (-1)\big[0 - \tfrac{a}{c}(a-\lambda)\big] = 0.$$

Expand the first inner bracket:

$$(a-\lambda)(-c-\lambda) = \lambda^2 + (c-a)\lambda - ac.$$

Multiply by $-\lambda$:

$$-\lambda(\lambda^2 + (c-a)\lambda - ac) = -\lambda^3 - (c-a)\lambda^2 + ac\lambda.$$

Combine all three cofactor contributions:

$$-\lambda^3 - (c-a)\lambda^2 + ac\lambda + (-c-\lambda) + \tfrac{a}{c}(a-\lambda) = 0.$$

Expand the last bracket:

$$-\lambda^3 - (c-a)\lambda^2 + ac\lambda - c - \lambda + \tfrac{a^2}{c} - \tfrac{a\lambda}{c} = 0.$$

Drop $O(1/c)$ terms (a^2/c and $a\lambda/c$) since $c \gg 1$:

$$-\lambda^3 - (c-a)\lambda^2 + ac\lambda - c - \lambda = 0.$$

Collect λ terms:

$$-\lambda^3 - (c-a)\lambda^2 + (ac-1)\lambda - c = 0.$$

Multiply by -1:

$$\lambda^3 + (c-a)\lambda^2 - (ac-1)\lambda + c = 0. \tag{$*$}$$

Anticipate one large real root $\lambda_3 \approx -c$ from leading $c\lambda^2 + c$ balance. Factor ($*$) exactly:

$$\lambda^3 + (c-a)\lambda^2 - (ac-1)\lambda + c = (\lambda + c)(\lambda^2 - a\lambda + 1).$$

Check by expansion: $(\lambda+c)(\lambda^2-a\lambda+1) = \lambda^3 - a\lambda^2 + \lambda + c\lambda^2 - ac\lambda + c = \lambda^3 + (c-a)\lambda^2 + (1-ac)\lambda + c$. ✓

Set the quadratic factor to zero for in-plane eigenvalues:

$$\lambda^2 - a\lambda + 1 = 0.$$

Solve:

$$\lambda_{1,2} = \tfrac{a\pm\sqrt{a^2-4}}{2}.$$

The third eigenvalue is $\lambda_3 \approx -c$ (strongly contracting along z-like direction).

Behaviour as a function of a (with $\lambda_3 \approx -c < 0$ always strongly stable):

- $-2 < a < 0$: discriminant $a^2 - 4 < 0$, $\operatorname{Re}\lambda_{1,2} = a/2 < 0$, complex pair: *stable spiral* in xy-plane $\Rightarrow$ stable focus overall.

- $a < -2$: real negative eigenvalues: stable node in xy $\Rightarrow$ stable node overall.

- $a = 0$: $\operatorname{Re}\lambda_{1,2} = 0$ (centre in xy-plane): Hopf bifurcation marginally.

- $0 < a < 2$: $\operatorname{Re}\lambda_{1,2} = a/2 > 0$, complex conjugate pair: *unstable spiral* in xy-plane $\Rightarrow$ saddle-focus (two unstable spiral directions, one stable real).

- $a > 2$: real positive eigenvalues, unstable node in xy combined with stable z $\Rightarrow$ saddle.

- $a = \pm 2$: degenerate (repeated real eigenvalue).

$$\boxed{\lambda^2 - a\lambda + 1 = 0,\ \lambda_3 \approx -c;\ \text{Hopf bifurcation at } a = 0.}$$

(c) Trajectory behaviour with $a = 0.1$, $c = 10$

Problem. Explain the figure: spiralling outward in the xy plane, then sudden vertical excursion when x exceeds c.

Initialised at the origin: the inner fixed point at $(a^2/c, -a/c, a/c) \approx (0.001, -0.01, 0.01)$ is a saddle-focus with $\operatorname{Re}\lambda_{1,2} = a/2 = 0.05 > 0$; the trajectory spirals outwards in the (approximately horizontal) xy plane while $z \approx 0$.

When x exceeds $c = 10$, the z-equation $\dot{z} = a + z(x - c)$ has $x - c > 0$, so z grows exponentially: a fast vertical excursion out of the xy plane. Once z is large, the $-z$ term in $\dot{x}$ rapidly drives x back below c, after which $x - c < 0$ damps z exponentially back to small values. The trajectory re-enters the spiral region but at a different radius, producing a chaotic orbit alternating slow horizontal spiralling with rare, sharp vertical kicks.

(d) Symmetric part of J and attractor geometry

Problem. Show that eigenvectors of $\mathbf{J} + \mathbf{J}^T$ identify growth/decay directions; show that for the Rössler system with $a > 0$ there is at least one growing and one shrinking direction at every point; interpret $\overline{\tau(\mathbf{J})} < 0$.

Energy of a perturbation $\delta\mathbf{x}$:

$$E = \tfrac{1}{2}|\delta\mathbf{x}|^2.$$

Take time derivative:

$$\dot{E} = \delta\mathbf{x} \cdot \delta\dot{\mathbf{x}}.$$

Linearise $\delta\dot{\mathbf{x}} = \mathbf{J}\,\delta\mathbf{x}$:

$$\dot{E} = \delta\mathbf{x} \cdot \mathbf{J}\,\delta\mathbf{x}.$$

Symmetrise (only the symmetric part contributes to the quadratic form $\mathbf{v} \cdot M\mathbf{v}$):

$$\dot{E} = \tfrac{1}{2}\,\delta\mathbf{x} \cdot (\mathbf{J} + \mathbf{J}^T)\,\delta\mathbf{x}.$$

The symmetric matrix $\mathbf{J} + \mathbf{J}^T$ is diagonalisable with real eigenvalues μ_i and orthonormal eigenvectors $\hat{\mathbf{e}}_i$. Decompose $\delta\mathbf{x} = \sum_i c_i\hat{\mathbf{e}}_i$:

$$\dot{E} = \tfrac{1}{2}\sum_i \mu_i c_i^2.$$

Along $\hat{\mathbf{e}}_i$ with $\mu_i > 0$, $|\delta\mathbf{x}|^2$ grows; with $\mu_i < 0$, it shrinks. $\square$

Compute $\mathbf{J} + \mathbf{J}^T$ for the Rössler system:

$$\mathbf{J} + \mathbf{J}^T = \begin{pmatrix} 0 & 0 & z-1 \\ 0 & 2a & 0 \\ z-1 & 0 & 2(x-c) \end{pmatrix}.$$

The y-axis decouples: $\hat{\mathbf{y}}$ is an eigenvector with eigenvalue $\mu_y = 2a$. For $a > 0$, $\mu_y > 0$, so perturbations along $\hat{\mathbf{y}}$ grow at every phase-space point. $\square_1$

The remaining xz-block of $\mathbf{J} + \mathbf{J}^T$ is

$$M_{xz} = \begin{pmatrix} 0 & z-1 \\ z-1 & 2(x-c) \end{pmatrix}.$$

Compute its determinant:

$$\det M_{xz} = 0 \cdot 2(x-c) - (z-1)^2 = -(z-1)^2.$$

Product of the two eigenvalues equals the determinant:

$$\mu_x \, \mu_z = -(z-1)^2 \le 0.$$

Therefore, generically (whenever $z \neq 1$), one of μ_x, μ_z is positive and the other negative: there is at least one shrinking direction at every such phase-space point. $\square_2$

Geometric meaning of $\overline{\tau(\mathbf{J})} < 0$: by Liouville's theorem, the local rate of phase-space volume change along a trajectory is $\nabla \cdot \dot{\mathbf{x}} = \tau(\mathbf{J})$. Averaging:

$$\tfrac{1}{V}\tfrac{\mathrm{d}V}{\mathrm{d}t} = \overline{\tau(\mathbf{J})} < 0.$$

Phase-space volumes contract on average. Combined with stretching in at least one direction (Lyapunov exponent positive), the attractor is locally folded into thinner sheets while being stretched along trajectories: the Rössler attractor is a *strange attractor of fractal dimension strictly between 2 and 3*.

Question 4 — Langevin and laser trap

(a) Overdamped limit; $\alpha = 2D\gamma^2$

Problem. Use the overdamped Langevin equation to derive $\alpha = 2D\gamma^2$ from the noise correlator.

Take $m/\gamma \to 0$, i.e. drop the $m\ddot{\mathbf{r}}$ term:

$$\gamma \, \dot{\mathbf{r}}(t) = \mathbf{F}(t).$$

Integrate from 0 to t:

$$\mathbf{r}(t) - \mathbf{r}(0) = \tfrac{1}{\gamma} \int_0^t \mathbf{F}(t') \, \mathrm{d}t'.$$

Square one Cartesian component and take ensemble average:

$$\langle [r_i(t) - r_i(0)]^2 \rangle = \tfrac{1}{\gamma^2} \int_0^t \int_0^t \langle F_i(t_1) F_i(t_2) \rangle \, \mathrm{d}t_1 \, \mathrm{d}t_2.$$

Insert correlator $\langle F_i(t_1) F_i(t_2) \rangle = \alpha \, \delta(t_2 - t_1)$:

$$\langle [r_i(t) - r_i(0)]^2 \rangle = \tfrac{\alpha}{\gamma^2} \int_0^t \int_0^t \delta(t_2 - t_1) \, \mathrm{d}t_1 \, \mathrm{d}t_2.$$

Perform the t_2 integral (delta picks $t_2 = t_1$):

$$\langle [r_i(t) - r_i(0)]^2 \rangle = \tfrac{\alpha}{\gamma^2} \int_0^t \mathrm{d}t_1 = \tfrac{\alpha t}{\gamma^2}.$$

Sum over $i = x, y, z$ for the full 3D displacement:

$$\langle |\Delta \mathbf{r}|^2 \rangle = \tfrac{3\alpha t}{\gamma^2}.$$

Equate to Einstein $\langle |\Delta \mathbf{r}|^2 \rangle = 6Dt$:

$$\tfrac{3\alpha t}{\gamma^2} = 6Dt.$$

Cancel t and solve for α:

$$\boxed{\alpha = 2D\gamma^2}$$

(b) Equipartition: $D\gamma = k_B T$; numerical D, $v_{\mathbf{rms}}$, diffusion time

Problem. From the full Langevin equation derive $D\gamma = k_B T$; for a 6 nm protein of mass 166×10^{-24} kg in water $\eta = 10^{-3}$ Pa s at 293 K, find D, v_{rms}, and time to diffuse $\sim 1\,\mu$m.

Take dot product of the Langevin equation with $\mathbf{r}$:

$$m\,\mathbf{r} \cdot \ddot{\mathbf{r}} + \gamma\,\mathbf{r} \cdot \dot{\mathbf{r}} = \mathbf{r} \cdot \mathbf{F}.$$

Apply identities $\mathbf{r} \cdot \ddot{\mathbf{r}} = \frac{1}{2}\frac{\mathrm{d}^2}{\mathrm{d}t^2}|\mathbf{r}|^2 - |\dot{\mathbf{r}}|^2$ and $\mathbf{r} \cdot \dot{\mathbf{r}} = \frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}|\mathbf{r}|^2$:

$$\tfrac{m}{2}\tfrac{\mathrm{d}^2}{\mathrm{d}t^2}|\mathbf{r}|^2 - m|\dot{\mathbf{r}}|^2 + \tfrac{\gamma}{2}\tfrac{\mathrm{d}}{\mathrm{d}t}|\mathbf{r}|^2 = \mathbf{r} \cdot \mathbf{F}.$$

Take ensemble average:

$$\tfrac{m}{2}\tfrac{\mathrm{d}^2}{\mathrm{d}t^2}\langle|\mathbf{r}|^2\rangle - m\langle|\dot{\mathbf{r}}|^2\rangle + \tfrac{\gamma}{2}\tfrac{\mathrm{d}}{\mathrm{d}t}\langle|\mathbf{r}|^2\rangle = \langle \mathbf{r} \cdot \mathbf{F} \rangle.$$

At equal time $\langle \mathbf{r}(t) \cdot \mathbf{F}(t) \rangle = 0$ (causality: $\mathbf{r}(t)$ depends on $\mathbf{F}$ only at earlier times):

$$\tfrac{m}{2}\tfrac{\mathrm{d}^2}{\mathrm{d}t^2}\langle|\mathbf{r}|^2\rangle - m\langle|\dot{\mathbf{r}}|^2\rangle + \tfrac{\gamma}{2}\tfrac{\mathrm{d}}{\mathrm{d}t}\langle|\mathbf{r}|^2\rangle = 0.$$

Apply equipartition (3 translational dof):

$$m\langle|\dot{\mathbf{r}}|^2\rangle = 3k_B T.$$

In steady-state diffusion, $\frac{\mathrm{d}}{\mathrm{d}t}\langle|\mathbf{r}|^2\rangle = 6D$ (constant), so the second derivative vanishes:

$$0 - 3k_B T + \tfrac{\gamma}{2}(6D) = 0.$$

Simplify:

$$3\gamma D = 3k_B T.$$

Divide by 3:

$$\boxed{D\gamma = k_B T}$$

(Einstein relation.)

Numerical evaluation. Stokes drag for a sphere of radius $a = 3 \times 10^{-9}$ m (half the diameter):

$$\gamma = 6\pi\eta a.$$

Insert $\eta = 10^{-3}$ Pa·s and $a = 3 \times 10^{-9}$ m:

$$\gamma = 6\pi(10^{-3})(3 \times 10^{-9})\ \mathrm{kg\,s^{-1}}.$$

Evaluate:

$$\gamma \approx 5.65 \times 10^{-11}\ \mathrm{kg\,s^{-1}}.$$

Compute $k_B T$ at $T = 293$ K:

$$k_B T = (1.38 \times 10^{-23})(293) \approx 4.04 \times 10^{-21}\ \mathrm{J}.$$

Diffusion coefficient from Einstein relation:

$$D = \tfrac{k_B T}{\gamma}.$$

Substitute:

$$D = \tfrac{4.04 \times 10^{-21}}{5.65 \times 10^{-11}}\ \mathrm{m^2\,s^{-1}}.$$

Evaluate:

$$\boxed{D \approx 7.2 \times 10^{-11}\,\mathrm{m^2\,s^{-1}}}$$

RMS speed from 3D equipartition $\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}k_B T$:

$$v_{\mathrm{rms}} = \sqrt{\tfrac{3k_B T}{m}}.$$

Substitute $m = 1.66 \times 10^{-22}$ kg and $k_B T = 4.04 \times 10^{-21}$ J:

$$v_{\mathrm{rms}} = \sqrt{\tfrac{3(4.04 \times 10^{-21})}{1.66 \times 10^{-22}}}\ \mathrm{m\,s^{-1}}.$$

Compute the numerator:

$$3(4.04 \times 10^{-21}) = 1.212 \times 10^{-20}\ \mathrm{J}.$$

Divide:

$$\tfrac{1.212 \times 10^{-20}}{1.66 \times 10^{-22}} = 73.0\ \mathrm{m^2\,s^{-2}}.$$

Take the square root:

$$\boxed{v_{\mathrm{rms}} \approx 8.5\,\mathrm{m\,s^{-1}}}$$

E. coli typical length $L \sim 2\,\mu\mathrm{m}$. Diffusion time from $\langle |\Delta\mathbf{r}|^2\rangle = 6Dt$:

$$t \sim \tfrac{L^2}{6D}.$$

Substitute $L = 2 \times 10^{-6}$ m and $D = 7.2 \times 10^{-11}$ m^2/s:

$$t \sim \tfrac{(2 \times 10^{-6})^2}{6(7.2 \times 10^{-11})}\ \mathrm{s}.$$

Simplify numerator and denominator:

$$t \sim \tfrac{4 \times 10^{-12}}{4.32 \times 10^{-10}}\ \mathrm{s}.$$

Evaluate:

$$\boxed{t \approx 10^{-2}\,\mathrm{s} = 10\,\mathrm{ms}}$$

(c) Mean-square displacement, power spectrum, and trap stiffness

Problem. For a bead in a 1D harmonic trap of stiffness κ, compute $\langle x^2\rangle$ from the partition function, then the position power spectrum from the Langevin equation; explain how it determines κ.

Trap potential:

$$U(x) = \tfrac{1}{2}\kappa x^2.$$

1D classical partition function:

$$Z = \int_{-\infty}^{\infty} e^{-U(x)/(k_B T)}\,\mathrm{d}x = \int_{-\infty}^{\infty} e^{-\kappa x^2/(2k_B T)}\,\mathrm{d}x.$$

Use Gaussian integral $\int e^{-ax^2}\mathrm{d}x = \sqrt{\pi/a}$ with $a = \kappa/(2k_B T)$:

$$Z = \sqrt{\tfrac{2\pi k_B T}{\kappa}}.$$

Mean-squared displacement:

$$\langle x^2\rangle = \tfrac{1}{Z}\int_{-\infty}^{\infty} x^2 e^{-\kappa x^2/(2k_B T)}\,\mathrm{d}x.$$

Use $\int x^2 e^{-ax^2}\mathrm{d}x = \frac{1}{2a}\sqrt{\pi/a}$ with $a = \kappa/(2k_B T)$:

$$\int x^2 e^{-\kappa x^2/(2k_B T)}\,\mathrm{d}x = \tfrac{k_B T}{\kappa}\sqrt{\tfrac{2\pi k_B T}{\kappa}}.$$

Divide by Z:

$$\boxed{\langle x^2\rangle = \tfrac{k_B T}{\kappa}}$$

(equipartition).

Overdamped 1D Langevin equation in the trap:

$$\gamma\,\dot{x} + \kappa x = F(t), \qquad \langle F(t_1)F(t_2)\rangle = 4k_B T\gamma\,\delta(t_2 - t_1).$$

Fourier-transform with convention $x(t) = \int e^{-i\omega t}\tilde{x}(\omega)\,\mathrm{d}\omega/(2\pi)$, so $\partial_t \to -i\omega$:

$$(-i\omega\gamma + \kappa)\tilde{x}(\omega) = \tilde{F}(\omega).$$

Solve for $\tilde{x}$:

$$\tilde{x}(\omega) = \tfrac{\tilde{F}(\omega)}{\kappa - i\omega\gamma}.$$

PSD of F from Wiener–Khinchin (FT of δ-correlator):

$$S_F(\omega) = \int \langle F(t)F(0)\rangle e^{i\omega t}\,\mathrm{d}t = 4k_B T\gamma.$$

Apply $S_x(\omega) = |H(\omega)|^2 S_F(\omega)$ with transfer function $H(\omega) = 1/(\kappa - i\omega\gamma)$:

$$|H(\omega)|^2 = \tfrac{1}{\kappa^2 + \omega^2\gamma^2}.$$

Substitute:

$$S_x(\omega) = \tfrac{S_F}{\kappa^2 + \omega^2\gamma^2}.$$

Insert $S_F = 4k_B T\gamma$:

$$\boxed{S_x(\omega) = \tfrac{4k_B T\gamma}{\kappa^2 + \gamma^2\omega^2}}$$

This is a Lorentzian. Define the corner (half-power) frequency:

$$\omega_c = \kappa/\gamma.$$

ω_c is read off from the knee in $\log S_x$ vs $\log\omega$. With $\gamma = 6\pi\eta a$ known independently:

$$\kappa = \gamma\,\omega_c.$$

(d) Autocorrelation ODE

Problem. Show $\gamma\,\dot{G}_x(\tau) + \kappa\,G_x(\tau) = 0$ for $\tau > 0$.

Definition (1D component x):

$$G_x(\tau) = \langle x(t)\,x(t-\tau)\rangle.$$

Take the overdamped Langevin equation evaluated at time t:

$$\gamma\,\dot{x}(t) + \kappa\,x(t) = F(t).$$

Multiply through by $x(t-\tau)$:

$$\gamma\,\dot{x}(t)\,x(t-\tau) + \kappa\,x(t)\,x(t-\tau) = F(t)\,x(t-\tau).$$

Take ensemble average:

$$\gamma\,\langle\dot{x}(t)\,x(t-\tau)\rangle + \kappa\,\langle x(t)\,x(t-\tau)\rangle = \langle F(t)\,x(t-\tau)\rangle.$$

For $\tau > 0$, $x(t-\tau)$ depends only on $F(t')$ with $t' \le t-\tau < t$; the delta-correlator $\langle F(t)F(t')\rangle \propto \delta(t-t')$ vanishes for $t' < t$. Hence:

$$\langle F(t)\,x(t-\tau)\rangle = 0.$$

Apply stationarity $\langle x(s_1)x(s_2)\rangle = G_x(s_1 - s_2)$ and differentiate in the first slot:

$$\langle\dot{x}(s_1)\,x(s_2)\rangle = G_x'(s_1 - s_2).$$

Set $s_1 = t$, $s_2 = t - \tau$:

$$\langle\dot{x}(t)\,x(t-\tau)\rangle = \tfrac{\mathrm{d}G_x(\tau)}{\mathrm{d}\tau}.$$

Identify $\langle x(t)\,x(t-\tau)\rangle = G_x(\tau)$ in the second term. Substitute both back:

$$\boxed{\gamma\,\tfrac{\mathrm{d}G_x(\tau)}{\mathrm{d}\tau} + \kappa\,G_x(\tau) = 0 \quad (\tau > 0)}$$

Solve: $G_x(\tau) = G_x(0)\,e^{-\kappa\tau/\gamma}$. Initial value $G_x(0) = \langle x^2\rangle = k_BT/\kappa$:

$$G_x(\tau) = \tfrac{k_BT}{\kappa}\,e^{-\kappa\tau/\gamma}.$$

Exponential decay with timescale $\gamma/\kappa = 1/\omega_c$, consistent with the Lorentzian PSD (Wiener–Khinchin).